profound. It makes the strange claim that observers in relative motion will have different perceptions of distance and of time. This means, as we shall see, that identical wristwatches worn by two individuals in relative motion will tick at different rates and hence will not agree on the amount of time that elapses between chosen events. Special relativity demonstrates that this statement does not slander the accuracy of the wristwatches involved; rather, it is a true statement about time itself.

Similarly, observers in relative motion carrying identical tape measures will not agree on the lengths of distances measured. Again, this is not due to inaccuracies in the measuring devices or to errors in how they are used. The most accurate measuring devices in the world confirm that space and time—as measured by distances and durations—are not experienced identically by everyone. In the precise way delineated by Einstein, special relativity resolves the conflict between our intuition about motion and the properties of light, but there is a price: individuals who are moving with respect to each other will not agree on their observations of either space or time.

It has been almost a century since Einstein informed the world of his dramatic discovery, yet most of us still see space and time in absolute terms. Special relativity is not in our bones—we do not feel it. Its implications are not a central part of our intuition. The reason for this is quite simple: The effects of special relativity depend upon how fast one moves, and at the speeds of cars, planes, or even space shuttles, these effects are minuscule. Differences in perceptions of space and of time between individuals planted on the earth and those traveling in cars or planes do occur, but they are so small that they go unnoticed. However, were one to take a trip in a futuristic space vehicle traveling at a substantial fraction of light speed, the effects of relativity would become plainly obvious. This, of course, is still in the realm of science fiction. Nevertheless, as we shall discuss in later sections, clever experiments allow clear and precise observation of the relative properties of space and time predicted by Einstein’s theory.

To get a sense of the scales involved, imagine that the year is 1970 and big, fast cars are in. Slim, having just spent all his savings on a new Trans Am, goes with his brother Jim to the local drag strip to give the car the kind of test-drive forbidden by the dealer. After revving up